

Intrinsic Closed-Loop Practical Asymptotic Stability in Discrete-Time Reinforcement Learning

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Summary

Deep reinforcement learning (RL) has achieved impressive empirical success in complex continuous control tasks, many of which are fundamentally set-stabilization problems. To safely deploy learned feedback laws, existing methods often modify RL algorithms to enforce asymptotic stability, for example, by embedding Lyapunov constraints or applying external safety filters. While these approaches provide stability certificates, they alter the core RL objective and often require exact zero-error convergence, which is practically infeasible to guarantee when using neural network function approximators.

This paper offers a complementary perspective: while such algorithmic modifications are valuable for training-time safety during exploration, modifying the core RL objective is often unnecessary to guarantee the stability of the deployed policy. We prove that the unmodified discounted RL objective intrinsically guarantees uniform practical asymptotic stability for deterministic continuous control tasks. By adapting the discrete-time input-to-state stability (ISS) framework, we model the suboptimality induced by both the discount factor and the neural network approximation error as bounded disturbances. Our analysis demonstrates that near-optimal parameterized policies inherently steer the closed-loop system into a bounded neighborhood of the target set. The size of this neighborhood vanishes in the limit as the approximation error approaches zero and, subsequently, the discount factor approaches one.

Contribution(s)

1. We prove that the exact optimal policy for the discounted reinforcement learning objective intrinsically guarantees closed-loop uniform practical asymptotic stability (UPAS) in deterministic continuous control tasks.

Context: Prior work typically enforces stability by modifying the RL algorithm (for example, via Lyapunov constraints (Han et al., 2020; Osinenko et al., 2020)) or applying external safety filters (Berkenkamp et al., 2017). While these modifications are valuable for training-time safety during exploration, our analysis evaluates the unmodified RL objective, demonstrating that modifying the core objective is often unnecessary to guarantee the stability of the deployed policy.

2. We extend this practical stability guarantee to near-optimal parameterized policies, proving that the ultimate steady-state error of the closed-loop system is explicitly bounded by the discount factor and the function approximation error.

Context: The discrete-time input-to-state stability (ISS) framework (Jiang & Wang, 2001) has been used to analyze the effect of discounting in model predictive control (MPC) (Postoyan et al., 2016). We extend this approach to the deep RL setting.

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Abstract

Deep reinforcement learning (RL) has achieved impressive empirical success in complex control tasks, many of which are fundamentally *set-stabilization* problems. To safely deploy learned policies, existing methods often modify RL algorithms to enforce strict asymptotic stability. However, these algorithmic modifications alter the core RL objective, and exact zero-error convergence is practically impossible to guarantee when using neural network function approximators. This paper offers a complementary perspective: while such algorithmic modifications are valuable for training-time safety during exploration, modifying the core RL objective is often unnecessary to guarantee the stability of the deployed policy. We prove that the unmodified discounted RL objective intrinsically guarantees closed-loop *practical asymptotic stability* in deterministic continuous control settings. By adapting the discrete-time input-to-state stability (ISS) framework and drawing inspiration from discounted model predictive control (MPC), we model the suboptimality induced by the discount factor and the neural network approximation error as bounded disturbances. We establish that near-optimal parameterized feedback laws steer the closed-loop system into a bounded neighborhood of the target set. The size of this neighborhood vanishes in the limit as the approximation error approaches zero and, subsequently, the discount factor approaches one.

1 Introduction

Deep reinforcement learning (RL) has empirically demonstrated impressive results in complex control tasks, such as dynamic legged locomotion (Hwangbo et al., 2019), acrobatic maneuvers (Peng et al., 2018), high-speed autonomous flight (Kaufmann et al., 2023), and dexterous manipulation (Andrychowicz et al., 2020). Fundamentally, these tasks are *set-stabilization* problems. Rather than driving a system to a single static equilibrium point, these tasks require the agent to converge to and indefinitely remain within a geometric *target set* in the state space. For example, dynamic legged locomotion requires stabilizing a continuous, repeating motion (a limit cycle). Similarly, autonomous drone racing involves converging to a spatial racing line, while dexterous manipulation requires steering an object into a manifold of valid target poses. While RL empirically excels at these tasks, formally guaranteeing closed-loop stability properties native to control theory remains challenging. Providing such formal guarantees is essential for safely deploying agents in real-world, safety-critical applications, such as autonomous driving or human-robot interaction.

To address this, a prominent branch of the literature modifies the RL objective to explicitly enforce formal stability guarantees. Many approaches target strict *asymptotic stability*, which guarantees that the closed-loop system eventually reaches and maintains a target behavior. Such approaches include jointly training neural Lyapunov functions (Richards et al., 2018; Chang et al., 2019; Han et al., 2020; Dai et al., 2021) or applying safety filters (Berkenkamp et al., 2017). Other methods relax this

requirement to practical stability, but still achieve it by embedding Lyapunov decrease constraints into the learning process (Osinenko et al., 2020; Han et al., 2021). While these techniques lead to asymptotic stability-like properties, they fundamentally alter the core RL objective, which can restrict natural exploration and complicate training.

In this work, we offer a complementary perspective: while such algorithmic modifications are valuable for training-time safety during exploration, modifying the core RL objective is often unnecessary to guarantee the stability of the deployed policy. We argue that *practical asymptotic stability*—where a system eventually enters and indefinitely remains within a small, bounded neighborhood of the target—is an intrinsic property of the near-optimal solution to the RL problem. In classical control theory, asymptotic stability is an infinite-horizon property. In contrast, RL optimizes an expected *discounted* return. As established in the discounted model predictive control (MPC) literature, see e.g., (Postoyan et al., 2016), a discount factor effectively truncates the optimization horizon. This yields a myopic agent that prioritizes near-term performance and mathematically settles for being “close enough” to the target. Thus, the native steady-state error observed in RL is not an algorithmic failure, but the direct consequence of the discounted objective.

To formally establish these intrinsic guarantees, we adapt the discrete-time input-to-state stability (ISS) framework in Jiang & Wang (2001) and draw inspiration from discounted MPC. While MPC typically relies on exact online optimization, practical RL uses parameterized neural networks that produce *near-optimal* solutions at best (Kidger & Lyons, 2020). Using ISS, we bridge this gap by treating both the truncated horizon (due to discounting) and the suboptimality of the agent (due to function approximation) as bounded disturbances to the ideal closed-loop system. To build this theoretical foundation, we first establish essential properties of the RL problem, including the continuity of the value functional and the uniform practical asymptotic stability of the exact optimal feedback law. We then extend our analysis to the approximate setting and derive near-optimality bounds for parameterized neural network-based feedback laws. Integrating these results, we prove that deep RL intrinsically provides practical closed-loop stability guarantees.

The remainder of this paper is organized as follows. Section 2.2 introduces the necessary notation and mathematical preliminaries. Section 3 formulates the ideal discrete-time RL problem and proves that the exact optimal feedback law intrinsically guarantees uniform practical asymptotic stability. Section 4 extends these stability guarantees to the practical approximate RL setting, where parameterized neural networks are used. Finally, Section 5 provides concluding remarks. For the sake of readability, all mathematical proofs are deferred to the supplementary material.

2 Preliminaries and Notation

2.1 Notation

The following notation is used throughout the paper. The n -dimensional Euclidean space is denoted by $\mathbb{R}^n$. The positive real numbers are denoted by $\mathbb{R}_{>0}$, i.e., $\mathbb{R}_{>0} := (0, \infty)$. The nonnegative real numbers are denoted by $\mathbb{R}_{\geq 0}$, i.e., $\mathbb{R}_{\geq 0} := [0, \infty)$. The natural numbers including 0 are denoted by $\mathbb{N}$, i.e., $\mathbb{N} := \{0, 1, 2, \dots\}$. The natural numbers excluding 0 are denoted by $\mathbb{N}_{>0}$, i.e., $\mathbb{N}_{>0} := \{1, 2, \dots\}$. The Euclidean norm of the vector x is denoted by $|x|$. The distance from x to the nonempty set S is denoted by $|x|_S$, which is given by $\inf_{y \in S} |x - y|$. A sequence of vectors $v_0, v_1, \dots$ is denoted by $\{v_k\}_{k \in \mathbb{N}}$.

2.2 Preliminaries

Definition 2.1 (Continuous Functions). *Let $\mathcal{X}$ be a compact subset of $\mathbb{R}^n$ and $\mathcal{U}$ a compact subset of $\mathbb{R}^m$. We denote by $\mathcal{C}(\mathcal{X})$ the space of all real-valued continuous functions on $\mathcal{X}$, endowed with the supremum norm*

$$|v|_{\infty} := \sup_{x \in \mathcal{X}} |v(x)|.$$

Similarly, $\mathcal{C}(\mathcal{X}, \mathcal{U})$ denotes the space of all continuous functions $\kappa: \mathcal{X} \rightarrow \mathcal{U}$ endowed with the supremum norm

$$|\kappa|_\infty := \sup_{x \in \mathcal{X}} |\kappa(x)|.$$

Equipped with the supremum norm, $\mathcal{C}(\mathcal{X})$ and $\mathcal{C}(\mathcal{X}, \mathbb{R}^m)$ are Banach spaces, and $\mathcal{C}(\mathcal{X}, \mathcal{U})$ is a complete metric space¹; see, for example, Theorem 7.15 in [Rudin \(1976\)](#). Moreover, $\mathcal{X}$ and $\mathcal{U}$ are metric spaces endowed with the supremum norm, inherited from $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively.

Definition 2.2 (Compactness in Function Space). *A subset $\mathbb{K} \subset \mathcal{C}(\mathcal{X}, \mathcal{U})$ is said to be compact with respect to the supremum norm if it is sequentially compact. That is, for every sequence $\{\kappa_n\}_{n \in \mathbb{N}}$ contained in $\mathbb{K}$, there exists a subsequence $\{\kappa_{n_j}\}_{j \in \mathbb{N}}$ and a limit function $\kappa^* \in \mathbb{K}$ such that*

$$\lim_{j \rightarrow \infty} |\kappa_{n_j} - \kappa^*|_\infty = 0.$$

Remark. An example of such a compact function space is the class of feedforward neural networks with a fixed architecture, continuous activation functions, and bounded weights, which are commonly used to parameterize continuous control policies in deep RL ([Peng et al., 2018](#); [Hwangbo et al., 2019](#); [Kaufmann et al., 2023](#)).

Definition 2.3 (Class- $\mathcal{K}$ functions). *A function $\alpha: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a class- $\mathcal{K}$ function, also written $\alpha \in \mathcal{K}$, if α is zero at zero, continuous, and strictly increasing.*

Definition 2.4 (Class- $\mathcal{KL}$ functions). *A function $\beta: \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is a class- $\mathcal{KL}$ function, also written $\beta \in \mathcal{KL}$, if it is nondecreasing in its first argument, nonincreasing in its second argument, $\lim_{r \searrow 0} \beta(r, s) = 0$ for each $s \in \mathbb{R}_{\geq 0}$, and $\lim_{s \rightarrow \infty} \beta(r, s) = 0$ for each $r \in \mathbb{R}_{\geq 0}$.*

3 Discrete-time Reinforcement Learning

3.1 System Definition

We consider a discrete-time (DT) controlled dynamical system of the form

$$x^+ = g(x, u), \tag{1}$$

where $x \in \mathcal{X}$ denotes the state variable, with $\mathcal{X}$ being the state space, $x^+ \in \mathcal{X}$ is the next value of the state, $u \in \mathcal{U}$ is the control input, with $\mathcal{U}$ being the action space, and $g: \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ is the DT dynamics. A solution to the DT system in (1) is defined as follows.

Definition 3.1 (Solution to the DT System). *Given an initial condition $x_0 \in \mathcal{X}$ and an input sequence $\{u_k\}_{k \in \mathbb{N}}$, a solution to the DT system in (1) is the sequence $\{x_k\}_{k \in \mathbb{N}}$ defined recursively by $x_{k+1} = g(x_k, u_k)$ for each $k \in \mathbb{N}$.²*

Let $R: \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$ be a continuous function capturing the reward and let $\gamma \in (0, 1)$ represent the discount factor. The total discounted reward of the DT system in (1) is described by the *reward functional* defined as follows.

Definition 3.2 (Reward Functional). *Fix $\gamma \in (0, 1)$. For a given initial condition $x_0 \in \mathcal{X}$ and an input sequence $\mathbf{u} := \{u_k\}_{k \in \mathbb{N}}$ generating a solution $\{x_k\}_{k \in \mathbb{N}}$ of the DT system in (1), the reward functional³ is defined as*

$$J_\gamma(x_0, \mathbf{u}) := \sum_{k=0}^{\infty} \gamma^k R(x_k, u_k).$$

Next, we assign the control input to a feedback law, defined as follows.

¹The space $\mathcal{C}(\mathcal{X}, \mathcal{U})$ is not a Banach space as $\mathcal{U}$ is not a vector space.

²By definition, all solutions are *complete*, meaning solutions are defined for all future discrete time steps $k \in \mathbb{N}$.

³For each x_0 , the mapping $\mathbf{u} \mapsto J_\gamma(x_0, \mathbf{u})$ is called a *functional* because its argument is a function (or sequence) rather than a finite-dimensional vector.

Definition 3.3 (Feedback Law and Closed-Loop System). *A feedback law is a function $\kappa \in \mathcal{C}(\mathcal{X}, \mathcal{U})$. The DT controlled dynamical system in (1) subject to a feedback law κ is given by the DT closed-loop system*

$$x^+ = g(x, \kappa(x)). \quad (2)$$

Note that, for each feedback law $\kappa \in \mathcal{C}(\mathcal{X}, \mathcal{U})$ and initial condition $x_0 \in \mathcal{X}$, there exists a unique solution $\{x_k^\kappa\}_{k \in \mathbb{N}}$ to the DT closed-loop system in (2) by the fact that $g \circ \kappa$ is single valued. Furthermore, given $(x_0, \kappa) \in \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U})$, the total discounted reward of the DT closed-loop system in (2) is described by the *value functional* defined as follows.

Definition 3.4 (Value Functional under Feedback). *Fix $\gamma \in (0, 1)$. Given an initial condition $x_0 \in \mathcal{X}$ and a feedback law $\kappa \in \mathcal{C}(\mathcal{X}, \mathcal{U})$ generating a solution $\{x_k^\kappa\}_{k \in \mathbb{N}}$ to the DT closed-loop system in (2), the value functional $\mathcal{V}_\gamma : \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U}) \rightarrow \mathbb{R}$ is defined as*

$$\mathcal{V}_\gamma(x_0, \kappa) := \sum_{k=0}^{\infty} \gamma^k R(x_k^\kappa, \kappa(x_k^\kappa)). \quad (3)$$

We make the following assumption on the DT dynamics g , the reward function R , and the state and action spaces $\mathcal{X}$ and $\mathcal{U}$.

Assumption 3.1. *The functions $g : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ and $R : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$ are continuous, and the sets $\mathcal{X}$ and $\mathcal{U}$ are compact.*

Remark. *In physical applications such as robotics, assuming the state and action spaces $\mathcal{X}$ and $\mathcal{U}$ are compact sets is not restrictive. In particular, workspaces are often bounded, and actuators have physical saturation limits. Furthermore, the DT dynamics g are typically obtained by discretizing continuous-time models. Because the underlying continuous-time dynamics of many physical systems, such as robotic manipulators and wheeled mobile robots, are typically smooth, discretizing their dynamics over a compact domain inherently yields a continuous mapping g . Lastly, a common choice of the reward function R is the negated Euclidean distance to a target set, which is a continuous function.*

3.2 Regularity of the Value Functional

In this section, useful properties of the value functional in (3) are derived. Establishing these properties is necessary to guarantee the existence of an optimal feedback law in Section 3.3 and to relate expected performance to pointwise error bounds when evaluating feedforward neural network-based feedback laws in Section 4.

First, we characterize the regularity of the closed-loop trajectories. Under Assumption 3.1, the evaluation map $(\kappa, x) \mapsto \kappa(x)$ is jointly continuous. Namely, for each $(\kappa, x) \in \mathcal{C}(\mathcal{X}, \mathcal{U}) \times \mathcal{X}$ and any $\varepsilon > 0$, there exists a $\delta > 0$ such that for any perturbed feedback law $\tilde{\kappa} \in \mathcal{C}(\mathcal{X}, \mathcal{U})$ and perturbed state $\tilde{x} \in \mathcal{X}$ satisfying $|\tilde{\kappa} - \kappa|_\infty < \delta$ and $|\tilde{x} - x| < \delta$, the resulting variation in the control input satisfies $|\tilde{\kappa}(\tilde{x}) - \kappa(x)| < \varepsilon$. Consequently, solutions to the DT closed-loop system in (2) depend continuously on the initial condition and the feedback law (see Lemmas 5.1 and 5.2 in the supplementary material).

This continuous dependence on the initial condition and the feedback law, along with the compactness of $\mathcal{X}$ and $\mathcal{U}$, the continuity of the reward function, and the discount factor, is sufficient to guarantee the continuity of the value functional, as we show next.

Theorem 3.2 (Continuity of the Value Functional). *Suppose Assumption 3.1 holds. Then, for each $\gamma \in (0, 1)$, the value functional $\mathcal{V}_\gamma : \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U}) \rightarrow \mathbb{R}$ is continuous with respect to the supremum norm.*

Lastly, the following lemma confirms that the value functional satisfies the Bellman equation. This recursive property is used in the subsequent stability analysis to evaluate the difference of the value functional along solutions to the closed-loop system.

Lemma 3.3 (Bellman Equation). *For each $\gamma \in (0, 1)$, the value functional $\mathcal{V}_\gamma: \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U}) \rightarrow \mathbb{R}$ satisfies the Bellman equation*

$$\mathcal{V}_\gamma(x_0, \kappa) = R(x_0, \kappa(x_0)) + \gamma \mathcal{V}_\gamma(g(x_0, \kappa(x_0)), \kappa), \quad (4)$$

for each $(x_0, \kappa) \in \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U})$.

3.3 Reinforcement Learning Formulation

In this section, we formulate the RL problem and prove the existence of an optimal feedback law. We restrict the optimization to a specific class of feedback laws, defined as follows.

Definition 3.5 (Admissible Feedback Laws). *A set of feedback laws $\mathbb{K} \subset \mathcal{C}(\mathcal{X}, \mathcal{U})$ is called admissible if it is a non-empty subset that is compact with respect to the supremum norm.*

Because the value functional $\mathcal{V}_\gamma$ depends on the initial state $x \in \mathcal{X}$, evaluating the global performance of a feedback law requires aggregating these pointwise values into a single scalar objective. To perform this aggregation, we define an *admissible* initial state distribution to weight the relative importance of each initial state.

Definition 3.6 (Admissible Initial State Distribution). *A function $p_0: \mathcal{X} \rightarrow \mathbb{R}_{>0}$ is called an admissible initial state distribution if it is a continuous, strictly positive function that satisfies*

$$\int_{\mathcal{X}} p_0(x) dx = 1.$$

Problem 1 (Reinforcement Learning Problem). *Given $\gamma \in (0, 1)$, an admissible initial state distribution $p_0: \mathcal{X} \rightarrow \mathbb{R}_{>0}$, and an admissible set of feedback laws $\mathbb{K}$, the RL problem is defined as*

$$\max_{\kappa \in \mathbb{K}} \mathcal{J}_\gamma(\kappa), \quad (5)$$

where $\mathcal{J}_\gamma: \mathbb{K} \rightarrow \mathbb{R}$ is the expected value functional defined as

$$\mathcal{J}_\gamma(\kappa) := \int_{\mathcal{X}} \mathcal{V}_\gamma(x, \kappa) p_0(x) dx \quad \forall \kappa \in \mathbb{K}. \quad (6)$$

In Problem 1, we aim to find a feedback law by searching over the admissible set of feedback laws to maximize the expected value functional over the state space, weighting the value of each initial state by the initial state distribution.

Remark. While RL typically maximizes value functionals over stochastic system dynamics and stochastic feedback laws (Sutton & Barto, 2018), the objective functional $\mathcal{J}_\gamma$ in Problem 1 assumes deterministic dynamics and deterministic feedback laws. This deterministic framework aligns with practical continuous control: physical systems are typically deterministic, and the stochastic feedback laws used for exploration during training typically converge to deterministic feedback laws as they approach optimality. Under these conditions, the typical RL objective and $\mathcal{J}_\gamma$ share the same optimal solution. Consequently, analyzing the closed-loop properties of the maximizer of $\mathcal{J}_\gamma$ directly characterizes the behavior of RL in deterministic settings.

The following lemma shows that the expected value functional in Problem 1 is continuous.

Lemma 3.4 (Continuity of the Expected Value Functional). *Suppose Assumption 3.1 holds. Then, for each $\gamma \in (0, 1)$, the expected value functional $\mathcal{J}_\gamma: \mathbb{K} \rightarrow \mathbb{R}$ is continuous with respect to the supremum norm.*

To conclude this section, we show that there exists an optimal feedback law that solves Problem 1.

Theorem 3.5 (Existence of the Optimal Feedback Law). *Suppose Assumption 3.1 holds. Then, for each $\gamma \in (0, 1)$, there exists an optimal feedback law $\kappa_\gamma^* \in \mathbb{K}$ that solves Problem 1; that is,*

$$\kappa_\gamma^* \in \arg \max_{\kappa \in \mathbb{K}} \mathcal{J}_\gamma(\kappa). \quad (7)$$

3.4 Properties of the Optimal Feedback Law

In this section, we study the properties of the optimal feedback law that solves Problem 1. We start by defining the *value function* for a given feedback law (instead of a *functional*) for notational brevity, see Footnote 3.

Definition 3.7 (Value Function of a Fixed Feedback Law). *Fix $\gamma \in (0, 1)$. Given a fixed feedback law $\kappa \in \mathcal{C}(\mathcal{X}, \mathcal{U})$, its corresponding value function $V_\gamma^\kappa: \mathcal{X} \rightarrow \mathbb{R}$ is defined as*

$$V_\gamma^\kappa(x) := \mathcal{V}_\gamma(x, \kappa) \quad \forall x \in \mathcal{X}. \quad (8)$$

Next, we define the notion of *proper indicator* (Goebel et al., 2012, Definition 7.9).

Definition 3.8 (Proper Indicator). *Given a compact target set $\mathcal{A} \subset \mathcal{X}$, a continuous function $\sigma: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ is called a proper indicator of $\mathcal{A}$ on $\mathcal{X}$ if it is positive definite with respect to $\mathcal{A}$, that is, $\sigma(x) = 0$ if and only if $x \in \mathcal{A}$.*

Remark. *In the context of RL, a proper indicator quantifies how far the agent deviates from the desired behavior (cf. stability of a desired target set). This formulation naturally captures a wide variety of standard continuous control tasks. For example, whether the task involves classical point stabilization, navigating to a broader goal region, or converging to a limit cycle (e.g., a periodic walking gait), the target set represents the desired state or motion, and the proper indicator corresponds to the penalty engineered into the reward function to drive the system toward this target.*

Inspired by Grimm et al. (2005); Postoyan et al. (2016), we make the following assumptions on the reward function and the set of admissible feedback laws.

Assumption 3.6 (Control Properties). *Given a reward function $R: \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$, a compact target set $\mathcal{A} \subset \mathcal{X}$, and a proper indicator $\sigma: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ of $\mathcal{A}$ on $\mathcal{X}$, we assume the following properties hold:*

(i) **Strict Reward Penalty:** *There exists $\underline{\alpha} \in \mathcal{K}$ such that*

$$R(x, u) \leq -\underline{\alpha}(\sigma(x)) \quad \forall (x, u) \in \mathcal{X} \times \mathcal{U}.$$

(ii) **Stabilizing Baseline:** *There exists $\kappa_s \in \mathbb{K}$ for which there exists a class- $\mathcal{K}$ function $\bar{\alpha}_V^{\kappa_s}$ such that, for each $\gamma \in (0, 1)$, its value function $V_\gamma^{\kappa_s}: \mathcal{X} \rightarrow \mathbb{R}$ satisfies*

$$-V_\gamma^{\kappa_s}(x) \leq \bar{\alpha}_V^{\kappa_s}(\sigma(x)) \quad \forall x \in \mathcal{X}.$$

The assumptions can be interpreted as follows. Item (i) in Assumption 3.6 ensures that the agent receives a strictly negative reward that decreases as the distance to the target set increases. For example, the negated Euclidean distance to a target set satisfies this condition. Item (ii) in Assumption 3.6 ensures that the stabilization task is solvable by a continuous feedback law, as illustrated in the following example.

Example 1. *Consider the DT system $x^+ = x + u$ with compact sets $\mathcal{X} = \mathcal{U} = [-1, 1]$. Let $\mathcal{A} = \{0\}$, $\sigma(x) = |x|$, and $R(x, u) = -(x^2 + u^2)$. Suppose $\mathbb{K}$ contains the proportional continuous feedback law $\kappa_s(x) = -0.5x$. Applying κ_s yields the closed-loop DT dynamics $x_{k+1} = 0.5x_k$ and the solution $x_k^{\kappa_s} = 0.5^k x_0$ for each $k \in \mathbb{N}$ and each initial state $x_0 \in \mathcal{X}$. For each $k \in \mathbb{N}$, $-R(x_k, u_k) = x_k^2 + (-0.5x_k)^2 = 1.25x_k^2$. Substituting in the solution $x_k^{\kappa_s}$ yields $-R(x_k, u_k) = 1.25((0.5)^k x_0)^2 = 1.25(0.25)^k x_0^2$. The negated value function for κ_s is then*

$$-V_\gamma^{\kappa_s}(x) = \sum_{k=0}^{\infty} \gamma^k (1.25(0.25)^k x^2) = 1.25x^2 \sum_{k=0}^{\infty} (0.25\gamma)^k \leq 1.25x^2 \sum_{k=0}^{\infty} (0.25)^k = \frac{5}{3}x^2,$$

where we use the fact that $\gamma \in (0, 1)$ and properties of the geometric series. By defining the class- $\mathcal{K}$ function $\bar{\alpha}_V^{\kappa_s}(s) := \frac{5}{3}s^2$, we obtain $-V_\gamma^{\kappa_s}(x) \leq \bar{\alpha}_V^{\kappa_s}(\sigma(x))$ for each $x \in \mathcal{X}$ and each $\gamma \in (0, 1)$.

Next, we make the following assumption on the existence of a pointwise optimal feedback law.

Assumption 3.7 (Ideal Expressivity). *For each $\gamma \in (0, 1)$, the admissible set of feedback laws $\mathbb{K}$ contains a pointwise optimal feedback law $\kappa_\gamma^\circ \in \mathbb{K}$, that is,*

$$V_\gamma^{\kappa_\gamma^\circ}(x) \geq V_\gamma^\kappa(x) \quad \forall (x, \kappa) \in \mathcal{X} \times \mathbb{K}.$$

To illustrate Assumption 3.7, we revisit Example 1.

Example 2. *Consider the system from Example 1. As the DT dynamics are linear and the cost quadratic, the pointwise optimal feedback law is of the form $\kappa_\gamma^\circ(x) = -K_\gamma x$, where K_γ is given by the discounted DT algebraic Riccati equation; see, for example, Section 3.2 in Bertsekas (2001). In particular, κ_γ° is continuous and K_γ is finite. Suppose $K_\gamma < K$, where $K > 0$ is finite. If we define $\mathbb{K}$ as the space of K -Lipschitz continuous functions, it naturally follows that $\kappa_\gamma^\circ \in \mathbb{K}$.*

Remark. *Assumption 3.7 implicitly assumes that Problem 1 can be pointwise optimally solved by a continuous feedback law. Consequently, this formulation excludes RL problems where the pointwise optimal feedback law is discontinuous, such as bang-bang control, and RL problems that contain topological obstructions, such as obstacle avoidance tasks (Priester et al., 2022; 2024). While this is a limitation of our theoretical work, discontinuous feedback laws typically lack robustness to measurement noise and execution delays, which can induce high-frequency chattering that degrades physical actuators. In fact, smooth continuous feedback laws are often preferred over aggressive, discontinuous optimal control in applications.*

The RL objective in Problem 1 evaluates policies by aggregating their return over the state space via an expected value functional. While maximizing an expected value generally guarantees optimality only in an averaged sense, the following lemma establishes a strictly stronger result. It proves that, under Assumption 3.7, the feedback law that maximizes the RL objective is exactly pointwise optimal for every individual state. Furthermore, we leverage this pointwise optimality to derive a uniform upper bound on the value function.

Lemma 3.8 (Pointwise Optimality of the Optimal Feedback Law). *Fix $\gamma \in (0, 1)$. Suppose Assumption 3.1, item (ii) in Assumption 3.6, and Assumption 3.7 hold. Let $\kappa_\gamma^* \in \mathbb{K}$ be an optimal feedback law that solves Problem 1. Then, its value function $V_\gamma^{\kappa_\gamma^*} : \mathcal{X} \rightarrow \mathbb{R}$ is pointwise optimal for each $x \in \mathcal{X}$, that is,*

$$V_\gamma^{\kappa_\gamma^*}(x) = V_\gamma^{\kappa_\gamma^\circ}(x) \quad \forall x \in \mathcal{X}, \forall \kappa_\gamma^\circ \in \mathbb{K}$$

Consequently, there exists $\kappa_s \in \mathbb{K}$ for which there exists a class- $\mathcal{K}$ function $\bar{\alpha}_V^{\kappa_s}$ such that, for each $\gamma \in (0, 1)$, the value function $V_\gamma^{\kappa_\gamma^} : \mathcal{X} \rightarrow \mathbb{R}$ satisfies*

$$-V_\gamma^{\kappa_\gamma^*}(x) \leq \bar{\alpha}_V^{\kappa_s}(\sigma(x)) \quad \forall x \in \mathcal{X}. \quad (9)$$

Building upon the uniform bound established in Lemma 3.8, we now present the main theoretical result of this section, which establishes that the DT closed-loop system in (2) subject to the optimal feedback law that solves Problem 1 renders the target set *uniformly practically asymptotically stable* (UPAS). This property guarantees that complete solutions to the closed-loop system approach and indefinitely remain within a bounded *neighborhood* around the target set.

Theorem 3.9 (Uniform Practical Asymptotic Stability of the Optimal Feedback Law). *Suppose Assumptions 3.1, 3.6, and 3.7 hold. Fix $\gamma \in (0, 1)$. Let $\sigma : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ be a proper indicator of $\mathcal{A}$ on $\mathcal{X}$ and $\kappa_\gamma^* \in \mathbb{K}$ denote the optimal feedback law that solves Problem 1, whose existence is guaranteed by Theorem 3.5. Then, there exist $\beta \in \mathcal{KL}$ and a continuous function $\nu : (0, 1) \rightarrow \mathbb{R}_{\geq 0}$ such that, for each initial state $x_0^* \in \mathcal{X}$, the solution $\{x_k^*\}_{k \in \mathbb{N}}$ to the DT closed-loop system in (2) generated by κ_γ^* satisfies*

$$\sigma(x_k^*) \leq \max \{ \beta(\sigma(x_0^*), k), \nu(\gamma) \} \quad \forall k \in \mathbb{N}, \quad (10)$$

where $\lim_{\gamma \rightarrow 1} \nu(\gamma) = 0$.

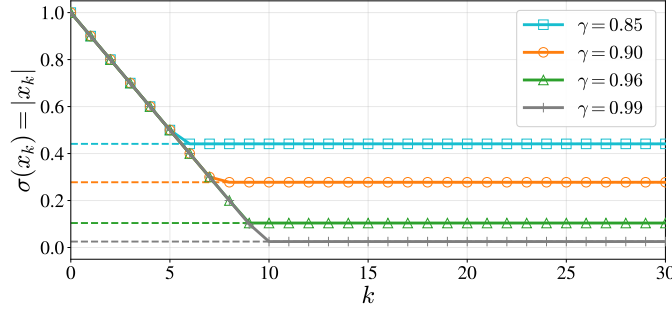


Figure 1: Solution trajectories generated by the optimal feedback law κ_γ^* to the closed-loop system defined in Example 3 for $\gamma \in \{0.85, 0.9, 0.96, 0.99\}$.

Importantly, the size of this neighborhood depends on the discount factor γ . In particular, a smaller discount factor corresponds to a myopic agent that can settle for a small steady-state error once it gets “close enough” to the target. As the agent becomes more farsighted, that is, as γ approaches 1, this neighborhood shrinks, and, in the limit, shrinks to the target set. The proof of Theorem 3.9 can be found in the supplementary material.

Example 3. Consider the DT system $x^+ = x + 0.1u$ on the compact domains $\mathcal{X} = [-1, 1]$ and $\mathcal{U} = [-1, 1]$. Let the target set be $\mathcal{A} = \{0\}$ with proper indicator $\sigma(x) = |x|$. To observe the ultimate bound $\nu(\gamma)$ established in Theorem 3.9, we define the reward function $R(x, u) = -(x^2 + 0.5|u|)$.

Because the linear control penalty dominates the quadratic state penalty near the origin, the optimal feedback law exhibits a steady-state error. Inside this ultimate bounding region, the optimal action is $u = 0$, as the immediate cost of a corrective step outweighs the discounted benefit of achieving a smaller future state penalty. In particular, the boundary of this steady-state error can be analytically determined as

$$\nu(\gamma) = \frac{5(1-\gamma)}{2\gamma}.$$

The optimal feedback law is given by the continuous function $\kappa_\gamma^*(x) = \text{clip}\left(-10 \max(x - \nu(\gamma), 0) + 10 \max(-x - \nu(\gamma), 0), -1, 1\right)$, where $\text{clip}(s, a, b) := \max\{a, \min\{s, b\}\}$ for each $(s, a, b) \in \mathbb{R}^3$. Figure 1 shows that the solution trajectories generated by κ_γ^* to the closed-loop system converge to their analytical boundaries $\nu(\gamma)$ for each $\gamma \in \{0.85, 0.9, 0.96, 0.99\}$. Furthermore, as γ increases, the steady-state error decreases.

As established in Theorem 3.9, the ultimate bound in (10) vanishes as the discount factor γ approaches one. However, in practice, pushing $\gamma \rightarrow 1$ often induces training instability. Fortunately, a direct and practical consequence of Theorem 3.9 is that, for any desired level of closeness, there exists a discount factor $\gamma < 1$ that is sufficiently large to achieve this tolerance without requiring the strict limit $\gamma \rightarrow 1$. For example, if we want to train a robotic manipulator to reach and stay within a particular distance to the target, we can always find a discount factor whose optimal feedback law guarantees this specification. The following corollary formalizes this property.

Corollary 3.9.1. Under the conditions of Theorem 3.9, for each $\rho > 0$, there exists $\gamma^* \in (0, 1)$ such that, for each $\gamma \in [\gamma^*, 1)$ and each initial state $x_0^* \in \mathcal{X}$, the solution $\{x_k^*\}_{k \in \mathbb{N}}$ to the DT closed-loop system in (2) generated by κ_γ^* satisfies

$$\sigma(x_k^*) \leq \max\{\beta(\sigma(x_0^*), k), \rho\} \quad \forall k \in \mathbb{N}. \quad (11)$$

4 Approximate Solution

Searching the infinite-dimensional space $\mathbb{K}$ to exactly solve Problem 1 is computationally intractable. Instead, RL algorithms typically restrict the search to a parameterized class of continuous feedback laws, specifically feedforward neural networks (Kidger & Lyons, 2020, Definition 3.1).

Definition 4.1 (Feedforward Neural Networks). *Let $\phi: \mathbb{R} \rightarrow \mathbb{R}$ and $n, m, w \in \mathbb{N}_{>0}$. Let $\mathcal{NN}_{n,m,w}^\phi$ represent the class of functions from $\mathbb{R}^n$ to $\mathbb{R}^m$ described by feedforward neural networks with n neurons in the input layer, m neurons in the output layer, and an arbitrary number of hidden layers, each with w neurons using the activation function ϕ . Every neuron in the output layer has the identity activation function.*

The following theorem and lemma guarantee that this feedforward neural network architecture is capable of uniformly approximating the value function of any continuous feedback law with arbitrary precision.

Theorem 4.1 (Universal Approximation (Kidger & Lyons, 2020, Theorem 3.2)). *Let $\phi: \mathbb{R} \rightarrow \mathbb{R}$ be any nonaffine continuous function that is continuously differentiable at at least one point, with a nonzero derivative at that point.⁴ Let $\mathcal{X} \subset \mathbb{R}^n$ be compact. Then, $\mathcal{NN}_{n,m,n+m+2}^\phi$ is dense in $\mathcal{C}(\mathcal{X}, \mathbb{R}^m)$ with respect to the supremum norm.*

Lemma 4.2 (Uniform Value Approximation). *Suppose Assumption 3.1 holds. Let $n, m \in \mathbb{N}_{>0}$ be the given dimensions of the state space $\mathcal{X}$ and action space $\mathcal{U}$, respectively, and let ϕ be an activation function satisfying the conditions of Theorem 4.1. Then, for each continuous feedback law $\kappa \in \mathbb{K}$ and tolerance $\delta > 0$, there exist $\ell \in \mathbb{N}_{>0}$ and a parameter vector $\bar{\psi} \in \mathbb{R}^\ell$ such that the parameterized network $\kappa_{\bar{\psi}} \in \mathcal{NN}_{n,m,n+m+2}^\phi \cap \mathbb{K}$ satisfies*

$$\max_{x \in \mathcal{X}} |V_\gamma^\kappa(x) - V_\gamma^{\kappa_{\bar{\psi}}}(x)| \leq \delta. \quad (12)$$

While Lemma 4.2 guarantees the existence of a near-optimal network, searching over all possible neural network architectures and parameters is computationally intractable. Instead, RL algorithms typically fix a specific architecture and optimize its finite-dimensional parameter vector, namely, weights and biases, over a bounded domain. We formalize this approximate RL problem as follows.

Problem 2 (Approximate Reinforcement Learning Problem). *Given a discount factor $\gamma \in (0, 1)$, state and action space dimensions $n, m \in \mathbb{N}_{>0}$, a feedforward neural network architecture $\mathcal{NN}_{n,m,w}^\phi$ with given activation function ϕ and hidden layer width $w \in \mathbb{N}_{>0}$, and a compact parameter set $\Psi \subset \mathbb{R}^\ell$ for some parameter dimension $\ell \in \mathbb{N}_{>0}$, the approximate RL problem is defined as*

$$\psi^* \in \arg \max_{\psi \in \Psi} \mathcal{J}_\gamma(\kappa_\psi) \quad \text{s.t.} \quad \kappa_\psi \in \mathcal{NN}_{n,m,w}^\phi \cap \mathbb{K}. \quad (13)$$

Remark. *Constraining the search to a compact parameter set Ψ ensures that the maximizers in Problem 2 are well-defined. Furthermore, in RL implementations, network parameters are typically bounded by regularization techniques, such as weight decay and gradient clipping.*

The following lemma guarantees that solving the approximate RL problem preserves near-optimal performance, provided the network architecture is chosen sufficiently large.

Lemma 4.3 (Near-Optimality of Approximate RL). *Suppose Assumption 3.1 holds. Let $n, m \in \mathbb{N}_{>0}$ be the given dimensions of the state space $\mathcal{X}$ and action space $\mathcal{U}$, respectively, and let ϕ be an activation function satisfying the conditions of Theorem 4.1. Then, for each $\kappa_\gamma^* \in \mathbb{K}$ that solves Problem 1 and each $\delta > 0$, there exist a parameter dimension $\ell \in \mathbb{N}_{>0}$ and a compact set $\Psi \subset \mathbb{R}^\ell$ such that an optimal parameterized feedback law $\kappa_{\psi^*} \in \mathcal{NN}_{n,m,n+m+2}^\phi \cap \mathbb{K}$ that solves Problem 2 using Ψ satisfies*

$$\mathcal{J}_\gamma(\kappa_{\psi^*}) \in [\mathcal{J}_\gamma(\kappa_\gamma^*) - \delta, \mathcal{J}_\gamma(\kappa_\gamma^*)] \quad (14)$$

⁴This condition is satisfied by almost all common activation functions. It includes smooth functions like tanh and sigmoid, as well as ReLU (since its derivative is strictly positive for all $x > 0$).

As formalized in Problems 1 and 2, RL algorithms maximize the expected return $\mathcal{J}_\gamma$, which evaluates the value functional integrated over the state space $\mathcal{X}$ for a given admissible state distribution. However, establishing UPAS requires a worst-case bound on the value function that holds pointwise across all individual states. By leveraging the compactness of the admissible set $\mathbb{K}$, the following lemma bridges this gap: it establishes that bounding the expected suboptimality of a feedback law, which is the metric optimized in Problem 2, guarantees a worst-case bound on its pointwise error over $\mathcal{X}$.

Lemma 4.4. *Fix $\gamma \in (0, 1)$. Suppose Assumption 3.1, item (ii) in Assumption 3.6, and Assumption 3.7 hold. Let $\kappa_\gamma^* \in \mathbb{K}$ denote an optimal feedback law that solves Problem 1, whose existence is guaranteed by Theorem 3.5. Then, there exists $\alpha_\gamma \in \mathcal{K}$ such that*

$$\max_{x \in \mathcal{X}} \left\{ V_\gamma^{\kappa_\gamma^*}(x) - V_\gamma^\kappa(x) \right\} \leq \alpha_\gamma(\mathcal{J}_\gamma(\kappa_\gamma^*) - \mathcal{J}_\gamma(\kappa)) \quad \forall \kappa \in \mathbb{K}. \quad (15)$$

We now extend the stability guarantees in Theorem 3.9 to the approximate RL of Problem 2. While computing the exact optimal policy is generally intractable, solving Problem 1 yields a parameterized feedback law with an approximation error δ . Our main theoretical result establishes that this parameterized feedback law still guarantees closed-loop UPAS. The ultimate steady-state error is now governed by the discount factor γ and the approximation error δ .

Theorem 4.5 (Uniform Practical Asymptotic Stability of the Optimal Parameterized Feedback Law). *Suppose Assumptions 3.1, 3.6, and 3.7 hold. Let $n, m \in \mathbb{N}_{>0}$ be the given dimensions of the state space $\mathcal{X}$ and action space $\mathcal{U}$, respectively, and let ϕ be an activation function satisfying the conditions of Theorem 4.1. Let $\kappa_\gamma^* \in \mathbb{K}$ denote an optimal feedback law that solves Problem 1, whose existence is guaranteed by Theorem 3.5. For each $\delta > 0$, let $\kappa_{\psi^*} \in \mathcal{NN}_{n,m,n+m+2}^\phi \cap \mathbb{K}$ denote an optimal parameterized feedback law, whose existence within the compact set Ψ for width $w = n + m + 2$ is guaranteed by Lemma 4.3, that solves Problem 2. Then, there exist $\beta \in \mathcal{KL}$ and a function $\tilde{\nu}: (0, 1) \times \mathbb{R}_{>0} \rightarrow \mathbb{R}_{\geq 0}$ such that for each $x_0 \in \mathcal{X}$, the solution $\{x_k\}_{k \in \mathbb{N}}$ to $x^+ = g(x, \kappa_{\psi^*}(x))$ satisfies*

$$\sigma(x_k) \leq \max \{ \beta(\sigma(x_0), k), \tilde{\nu}(\gamma, \delta) \} \quad \forall k \in \mathbb{N}, \quad (16)$$

where $\lim_{\gamma \rightarrow 1} (\lim_{\delta \rightarrow 0} \tilde{\nu}(\gamma, \delta)) = 0$.

Example 4. *Consider the setup from Example 3 with a fixed discount factor $\gamma = 0.90$. To observe the ultimate bound $\tilde{\nu}(\gamma, \delta)$ established in Theorem 4.5, we evaluate parameterized feedback laws κ_{ψ^*} subject to varying approximation errors $\delta > 0$. Testing these varying δ values can be thought of as simulating the RL training process, where the approximation error δ is progressively reduced as the network parameters are optimized.*

As established in Theorem 4.5, the approximation error δ acts as a bounded disturbance to the closed-loop system, strictly enlarging the steady-state error as δ increases. We model the parameterized policy $\kappa_{\psi^*} \in \mathcal{NN}_{n,m,w}^\phi$ as a two-neuron ReLU network given by

$$\kappa_{\psi^*}(x) = \text{clip} \left(-10 \max(w_\delta x - \nu(0.90), 0) + 10 \max(-w_\delta x - \nu(0.90), 0), -1, 1 \right).$$

Here, the approximation error δ is induced by a degraded network weight $w_\delta \in (0, 1]$. This weight artificially dilates the agent's perceived distance to the target, thereby expanding the steady-state error boundary from the analytical bound $\nu(0.90)$ to $\tilde{\nu}(\gamma, \delta) := \nu(0.90)/w_\delta$.

By numerically evaluating the value functions over the state space $\mathcal{X}$, we map the improving weights $w_\delta \in \{0.6, 0.8, 0.9\}$ to their maximum pointwise approximation errors $\delta = \max_{x \in \mathcal{X}} |V_\gamma^{\kappa_\gamma^*}(x) - V_\gamma^{\kappa_{\psi^*}}(x)|$, yielding a sequence of shrinking errors $\delta \in \{0.30, 0.04, 0.01\}$, respectively. As shown in Figure 2, the closed-loop trajectories converge to their boundaries $\tilde{\nu}(\gamma, \delta)$. Consistent with the practical stability bound in (16), larger approximation errors δ yield proportionally larger steady-state errors beyond the analytical boundary $\nu(\gamma)$.

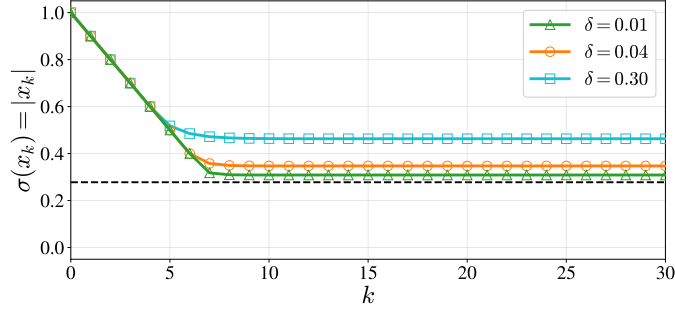


Figure 2: Solution trajectories generated by the parameterized feedback laws κ_{ψ^*} to the closed-loop system defined in Example 4 for $\gamma = 0.9$ and $\delta \in \{0.01, 0.04, 0.30\}$.

5 Conclusion

In this paper, we showed that RL intrinsically guarantees a practical notion of closed-loop stability for continuous control tasks under mild assumptions without requiring algorithmic modifications. By adapting the discrete-time input-to-state stability (ISS) framework, we modeled the suboptimality induced by the discount factor and the neural network approximation error as bounded disturbances. We proved that near-optimal parameterized policies inherently steer the closed-loop system into a bounded neighborhood of the target set. The size of this neighborhood vanishes in the limit as the approximation error approaches zero and, subsequently, the discount factor approaches one. Future work includes accommodating discontinuous optimal feedback laws to handle tasks with topological obstructions, such as obstacle avoidance. Additionally, while our current analysis is restricted to deterministic systems, we believe there are no fundamental hurdles to extending these proofs to stochastic dynamics; by incorporating probabilistic transitions into the framework, we hypothesize that similar ultimate bounds can be derived that hold in expectation.

Acknowledgments

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Supplementary Materials

The following content was not necessarily subject to peer review.

The following lemma shows that the map $(\kappa, x) \mapsto \kappa(x)$ is jointly continuous.

Lemma 5.1 (Continuity of the Evaluation Map). *Suppose the set $\mathcal{X}$ is compact and define the evaluation map*

$$\text{ev} : \mathcal{C}(\mathcal{X}, \mathcal{U}) \times \mathcal{X} \rightarrow \mathcal{U}$$

as

$$\text{ev}(\kappa, x) := \kappa(x) \quad \forall (\kappa, x) \in \mathcal{C}(\mathcal{X}, \mathcal{U}) \times \mathcal{X}.$$

Then, ev is jointly continuous.

Proof of Lemma 5.1. Fix $(\kappa, x) \in \mathcal{C}(\mathcal{X}, \mathcal{U}) \times \mathcal{X}$ and pick a sequence $\{(\kappa_k, x_k)\}_{k \in \mathbb{N}}$ such that

$$(\kappa_k, x_k) \in \mathcal{C}(\mathcal{X}, \mathcal{U}) \times \mathcal{X} \quad \forall k \in \mathbb{N}$$

and

$$\lim_{k \rightarrow \infty} (\kappa_k, x_k) = (\kappa, x).$$

By the definition of the product topology, this means

$$\lim_{k \rightarrow \infty} \|\kappa_k - \kappa\|_\infty = 0, \quad \lim_{k \rightarrow \infty} x_k = x.$$

We have

$$\begin{aligned} |\kappa_k(x_k) - \kappa(x)| &= |\kappa_k(x_k) - \kappa(x_k) + \kappa(x_k) - \kappa(x)| \\ &\leq |\kappa_k(x_k) - \kappa(x_k)| + |\kappa(x_k) - \kappa(x)|. \end{aligned}$$

For the first term, by the definition of the supremum norm,

$$|\kappa_k(x_k) - \kappa(x_k)| \leq \|\kappa_k - \kappa\|_\infty.$$

Then,

$$\lim_{k \rightarrow \infty} |\kappa_k(x_k) - \kappa(x_k)| \leq \lim_{k \rightarrow \infty} \|\kappa_k - \kappa\|_\infty = 0.$$

For the second term, since $\mathcal{X}$ is compact and κ is continuous, κ is uniformly continuous on $\mathcal{X}$ (see Theorem 4.19 in [Rudin \(1976\)](#)). By the sequential characterization of continuity,

$$\lim_{k \rightarrow \infty} x_k = x \implies \lim_{k \rightarrow \infty} \kappa(x_k) = \kappa(x).$$

Therefore,

$$\lim_{k \rightarrow \infty} |\kappa(x_k) - \kappa(x)| = 0.$$

Substituting the limits from the first and second terms back into the triangle inequality yields

$$\lim_{k \rightarrow \infty} |\kappa_k(x_k) - \kappa(x)| \leq \underbrace{\lim_{k \rightarrow \infty} |\kappa_k(x_k) - \kappa(x_k)|}_0 + \underbrace{\lim_{k \rightarrow \infty} |\kappa(x_k) - \kappa(x)|}_0 = 0.$$

Consequently,

$$\lim_{k \rightarrow \infty} \kappa_k(x_k) = \kappa(x),$$

which proves that the evaluation map ev is jointly continuous. $\square$

The following lemma shows that a solution to the DT closed-loop system in (2) depends continuously on the initial condition and the feedback law.

Lemma 5.2 (Continuous Dependence). *Suppose the function $g: \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ is continuous and the set $\mathcal{X}$ is compact. Then, for each $k \in \mathbb{N}$, the map $(\kappa, x_0) \mapsto x_k^\kappa$ from $\mathcal{C}(\mathcal{X}, \mathcal{U}) \times \mathcal{X}$ to $\mathcal{X}$, where x_k^κ is the solution to the DT closed-loop system in (2) at time step k generated by κ starting from x_0 , is continuous.*

Proof of Lemma 5.2. The result follows by induction on k . For $k = 0$, the map $(\kappa, x_0) \mapsto x_0$ is the projection onto the second component, which is trivially continuous. Assume continuity holds for k . Then

$$x_{k+1}^\kappa = g(x_k^\kappa, \kappa(x_k^\kappa)).$$

Lemma 5.1 holds as $\mathcal{X}$ is compact by assumption. Consequently, the evaluation map $(\kappa, x) \mapsto \kappa(x)$ is continuous on $\mathcal{C}(\mathcal{X}, \mathcal{U}) \times \mathcal{X}$ under the supremum norm, and g is continuous by assumption. Since x_k^κ is continuous by the inductive hypothesis, the composition is continuous. This proves the claim. $\square$

Proof of Theorem 3.2. Let $\gamma \in (0, 1)$ and $(x_0, \kappa) \in \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U})$ be given. Recall that $\{x_k^\kappa\}_{k \in \mathbb{N}}$ is the solution to the DT closed-loop system in (2) generated by κ starting from x_0 . Define the partial sum

$$\mathcal{V}_\gamma^N(x_0, \kappa) := \sum_{k=0}^N \gamma^k R(x_k^\kappa, \kappa(x_k^\kappa))$$

for some $N \in \mathbb{N}$. Under the assumptions that g is continuous and $\mathcal{X}$ compact, Lemma 5.2 guarantees that x_k^κ is jointly continuous in x_0 and κ . Furthermore, as R is assumed to be continuous, each term in the sum is continuous. Hence, the functional $\kappa \mapsto \mathcal{V}_\gamma^N$ is continuous as it is the finite sum of continuous functions. By assumption, $\mathcal{X}$ and $\mathcal{U}$ are compact, hence $\mathcal{X} \times \mathcal{U}$ is compact. Then, as R is continuous, there exists $M > 0$ such that $|R(x, u)| \leq M$ for each $(x, u) \in \mathcal{X} \times \mathcal{U}$ by Theorem 4.16 in Rudin (1976). We have the following bound: for each $(x_0, \kappa) \in \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U})$,

$$|\mathcal{V}_\gamma(x_0, \kappa) - \mathcal{V}_\gamma^N(x_0, \kappa)| \leq \sum_{k=N+1}^{\infty} \gamma^k |R(x_k^\kappa, \kappa(x_k^\kappa))| \leq \sum_{k=N+1}^{\infty} \gamma^k M = \frac{\gamma^{N+1}}{1-\gamma} M$$

where we use the fact that $\gamma \in (0, 1)$ and properties of the geometric series. The bound is uniform in x_0 and κ , implying that, as N approaches infinity, $\mathcal{V}_\gamma^N$ approaches $\mathcal{V}_\gamma$ uniformly. By Theorem 7.12 in Rudin (1976), the uniform limit of continuous functions is continuous. Hence, the functional $(x_0, \kappa) \mapsto \mathcal{V}_\gamma(x_0, \kappa)$ is continuous. $\square$

Proof of Lemma 3.3. Fix $(x_0, \kappa) \in \mathcal{X} \times \mathcal{C}(\mathcal{X}, \mathcal{U})$. By definition,

$$\mathcal{V}_\gamma(x_0, \kappa) = \sum_{k=0}^{\infty} \gamma^k R(x_k^\kappa, \kappa(x_k^\kappa)).$$

Separating the first term ($k = 0$) from the sum yields

$$\begin{aligned} \mathcal{V}_\gamma(x_0, \kappa) &= R(x_0^\kappa, \kappa(x_0^\kappa)) + \sum_{k=1}^{\infty} \gamma^k R(x_k^\kappa, \kappa(x_k^\kappa)) \\ &= R(x_0, \kappa(x_0)) + \gamma \sum_{k=1}^{\infty} \gamma^{k-1} R(x_k^\kappa, \kappa(x_k^\kappa)). \end{aligned}$$

We re-index the sum by letting $j = k - 1$. Then,

$$\begin{aligned} \mathcal{V}_\gamma(x_0, \kappa) &= R(x_0, \kappa(x_0)) + \gamma \sum_{j=0}^{\infty} \gamma^j R(x_{j+1}^\kappa, \kappa(x_{j+1}^\kappa)) \\ &= R(x_0, \kappa(x_0)) + \gamma \sum_{j=0}^{\infty} \gamma^j R(g(x_j^\kappa, \kappa(x_j^\kappa)), \kappa(g(x_j^\kappa, \kappa(x_j^\kappa)))) \\ &= R(x_0, \kappa(x_0)) + \gamma \mathcal{V}_\gamma(g(x_0, \kappa(x_0)), \kappa). \end{aligned}$$

This confirms that the Bellman equation holds. $\square$

Proof of Lemma 3.4. Under Assumption 3.1, Theorem 3.2 implies that for each $\gamma \in (0, 1)$, the mapping $(x, \kappa) \mapsto \mathcal{V}_\gamma(x, \kappa)$ is continuous with respect to the supremum norm. By assumption, $\mathcal{X}$ and $\mathbb{K}$ are compact metric spaces. Hence, the product $\mathcal{X} \times \mathbb{K}$ is a compact metric space.

Since the mapping $(x, \kappa) \mapsto \mathcal{V}_\gamma(x, \kappa)$ is continuous on the compact metric space $\mathcal{X} \times \mathbb{K}$, it is uniformly continuous by Theorem 4.19 in Rudin (1976). Thus, for any $\epsilon > 0$, there exists $\delta > 0$ such that for each $x \in \mathcal{X}$ and $\kappa_1, \kappa_2 \in \mathbb{K}$ satisfying $|\kappa_1 - \kappa_2|_\infty < \delta$, it holds that

$$|\mathcal{V}_\gamma(x, \kappa_1) - \mathcal{V}_\gamma(x, \kappa_2)| < \epsilon.$$

Then, for any such $\kappa_1, \kappa_2 \in \mathbb{K}$, the following bound holds for their difference in expected value:

$$\begin{aligned} |\mathcal{J}_\gamma(\kappa_1) - \mathcal{J}_\gamma(\kappa_2)| &= \left| \int_{\mathcal{X}} (\mathcal{V}_\gamma(x, \kappa_1) - \mathcal{V}_\gamma(x, \kappa_2)) p_0(x) dx \right| \\ &\leq \int_{\mathcal{X}} |\mathcal{V}_\gamma(x, \kappa_1) - \mathcal{V}_\gamma(x, \kappa_2)| p_0(x) dx \\ &< \int_{\mathcal{X}} \epsilon p_0(x) dx = \epsilon, \end{aligned}$$

where the final equality holds because p_0 is an admissible initial state distribution, which by definition satisfies $\int_{\mathcal{X}} p_0(x) dx = 1$. This proves that the functional $\kappa \mapsto \mathcal{J}_\gamma(\kappa)$ is continuous. $\square$

Proof of Theorem 3.5. Under Assumption 3.1, Lemma 3.4 guarantees that for each $\gamma \in (0, 1)$, the functional $\kappa \mapsto \mathcal{J}_\gamma(\kappa)$ is continuous on $\mathbb{K}$ with respect to the supremum norm. Since the admissible set of feedback laws $\mathbb{K}$ is a compact subset of a normed space by definition, the functional $\kappa \mapsto \mathcal{J}_\gamma(\kappa)$ attains its maximum on $\mathbb{K}$ by Theorem 1 in Luenberger (1997, Section 2.13), which results in the existence of κ_γ^* satisfying (7). $\square$

Proof of Lemma 3.8. By Assumption 3.7, $V_\gamma^{\kappa_\gamma^\circ}(x) \geq V_\gamma^{\kappa_\gamma^*}(x)$ for each $(x, \kappa) \in \mathcal{X} \times \mathbb{K}$. We define the non-negative error function $E: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ as

$$E(x) := V_\gamma^{\kappa_\gamma^\circ}(x) - V_\gamma^{\kappa_\gamma^*}(x) \geq 0 \quad \forall x \in \mathcal{X}.$$

Because κ_γ^* is defined as the maximizer of the expected value functional $\mathcal{J}_\gamma$ over $\mathbb{K}$, and $\kappa_\gamma^\circ \in \mathbb{K}$, it holds that $\mathcal{J}_\gamma(\kappa_\gamma^*) \geq \mathcal{J}_\gamma(\kappa_\gamma^\circ)$, and therefore $\mathcal{J}_\gamma(\kappa_\gamma^\circ) - \mathcal{J}_\gamma(\kappa_\gamma^*) \leq 0$. By the definition of the expected value functional, we obtain

$$\begin{aligned} \mathcal{J}_\gamma(\kappa_\gamma^\circ) - \mathcal{J}_\gamma(\kappa_\gamma^*) &= \int_{\mathcal{X}} \mathcal{V}_\gamma(x, \kappa_\gamma^\circ) p_0(x) dx - \int_{\mathcal{X}} \mathcal{V}_\gamma(x, \kappa_\gamma^*) p_0(x) dx \\ &= \int_{\mathcal{X}} (V_\gamma^{\kappa_\gamma^\circ}(x) - V_\gamma^{\kappa_\gamma^*}(x)) p_0(x) dx \\ &= \int_{\mathcal{X}} E(x) p_0(x) dx \leq 0. \end{aligned}$$

Under Assumption 3.1, Theorem 3.2 establishes that the mapping $(x, \kappa) \mapsto \mathcal{V}_\gamma(x, \kappa)$ is continuous and $x \mapsto V_\gamma^\kappa(x)$ is continuous for any fixed $\kappa \in \mathbb{K}$. Therefore, $E: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ is continuous as well. Because both $E(x) \geq 0$ and $p_0(x) > 0$ for each $x \in \mathcal{X}$, their product $x \mapsto E(x)p_0(x)$ is a non-negative continuous function. Since the integral of any non-negative function is non-negative, and $\int_{\mathcal{X}} E(x)p_0(x) dx \leq 0$, it must hold that $\int_{\mathcal{X}} E(x)p_0(x) dx = 0$. Furthermore, because the integral of a non-negative continuous function evaluates to zero if and only if the integrand is identically zero, we conclude that $E(x)p_0(x) = 0$ for each $x \in \mathcal{X}$. By definition of an admissible initial state

distribution, $p_0(x)$ is strictly positive for each $x \in \mathcal{X}$. Hence, we conclude that $E(x) = 0$ for each $x \in \mathcal{X}$. Consequently, $V_{\gamma}^{\kappa_{\gamma}^*}(x) = V_{\gamma}^{\kappa_{\gamma}^{\circ}}(x)$ for each $x \in \mathcal{X}$.

Lastly, by Assumption 3.7, and by the pointwise optimality of $V_{\gamma}^{\kappa_{\gamma}^*}$ proven above, it holds that

$$V_{\gamma}^{\kappa_{\gamma}^{\circ}}(x) = V_{\gamma}^{\kappa_{\gamma}^*}(x) \geq V_{\gamma}^{\kappa_s}(x) \quad \forall x \in \mathcal{X}.$$

Then, by item (ii) in Assumption 3.6, we obtain

$$-V_{\gamma}^{\kappa_{\gamma}^*}(x) \leq -V_{\gamma}^{\kappa_s}(x) \leq \bar{\alpha}_{V^s}(\sigma(x)) \quad \forall x \in \mathcal{X}.$$

□

Proof of Theorem 3.9. To establish the UPAS bound in (10), we use the DT input-to-state stability (ISS) framework from Jiang & Wang (2001). Although the DT closed-loop system $x^+ = g(x, \kappa_{\gamma}^*(x))$ is autonomous, the suboptimality induced by the discount factor $\gamma \in (0, 1)$ can be treated as a constant external *disturbance*. This allows us to map our setup to the generic perturbed DT system $x^+ = f(x, u)$ defined in Jiang & Wang (2001, Eq. (2)), where the disturbance input is a constant sequence satisfying $|u| \equiv d_{\gamma} \geq 0$.

If $x^+ = f(x, u)$ is ISS with respect to the disturbance d_{γ} , the solutions $\{x_k^*\}_{k \in \mathbb{N}}$ to the DT closed-loop system in (2) generated by κ_{γ}^* satisfy the bound (Jiang & Wang, 2001, Eq. (21))

$$\sigma(x_k^*) \leq \max \{ \beta(\sigma(x_0^*), k), \nu_{\text{ISS}}(d_{\gamma}) \},$$

for some $\beta \in \mathcal{KL}$ and $\nu_{\text{ISS}} \in \mathcal{K}$. By defining the ultimate residual bound as $\nu(\gamma) := \nu_{\text{ISS}}(d_{\gamma})$, this recovers our target UPAS bound in (10).

To prove that $x^+ = f(x, u)$ is indeed ISS, we apply Lemma 3.5 in Jiang & Wang (2001), which states that a system is ISS if it admits a valid ISS-Lyapunov function. In particular, we need to prove that there exist a continuous function $Y: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$, functions $\underline{\alpha}_Y, \bar{\alpha}_Y, \alpha_Y \in \mathcal{K}$, and a constant $d_{\gamma} \geq 0$ such that, for each $x \in \mathcal{X}$,

$$\underline{\alpha}_Y(\sigma(x)) \leq Y(x) \leq \bar{\alpha}_Y(\sigma(x)) \quad (17a)$$

$$Y(g(x, \kappa_{\gamma}^*(x))) - Y(x) \leq -\alpha_Y(Y(x)) + d_{\gamma} \quad (17b)$$

For brevity, given $\gamma \in (0, 1)$ and $R: \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$, let $V^*(x) := V_{\gamma}^{\kappa_{\gamma}^*}(x)$ and $R^*(x) := R(x, \kappa_{\gamma}^*(x))$ for each $x \in \mathcal{X}$. We define the composite Lyapunov candidate $Y(x) := -V^*(x)$.

First, we show that the lower bound in (17a) holds. By item (i) in Assumption 3.6, there exists $\underline{\alpha} \in \mathcal{K}$ such that $R(x, u) \leq -\underline{\alpha}(\sigma(x))$ for each $(x, u) \in \mathcal{X} \times \mathcal{U}$. Because $\underline{\alpha} \in \mathcal{K}$ and the proper indicator takes nonnegative values, it holds that $\underline{\alpha}(\sigma(x)) \geq 0$ for each $x \in \mathcal{X}$, which implies $R(x, u) \leq 0$ for each $(x, u) \in \mathcal{X} \times \mathcal{U}$. Let $\{x_k^*\}_{k \in \mathbb{N}}$ be the solution to the DT closed-loop system in (2) generated by κ_{γ}^* starting from the initial condition $x_0^* = x$. By the definition of V^* , we can separate the initial reward term to obtain

$$\begin{aligned} V^*(x) &= \sum_{k=0}^{\infty} \gamma^k R^*(x_k^*) \\ &= R^*(x) + \sum_{k=1}^{\infty} \gamma^k R^*(x_k^*) \end{aligned}$$

Because $\gamma \in (0, 1)$ and $R(x, u) \leq 0$ for each $(x, u) \in \mathcal{X} \times \mathcal{U}$, $\sum_{k=1}^{\infty} \gamma^k R^*(x_k^*) \leq 0$. Thus, for each $x \in \mathcal{X}$, $V^*(x)$ is upper-bounded by its initial reward term

$$V^*(x) \leq R^*(x) \leq -\underline{\alpha}(\sigma(x)).$$

Multiplying by -1 yields $Y(x) = -V^*(x) \geq \underline{\alpha}(\sigma(x)) =: \underline{\alpha}_Y(\sigma(x))$ for each $x \in \mathcal{X}$.

Furthermore, under the Assumptions 3.1, 3.6, and 3.7, Lemma 3.8 yields $Y(x) = -V^*(x) \leq \overline{\alpha}_{V^*}^{\kappa_s}(\sigma(x)) =: \overline{\alpha}_Y(\sigma(x))$ for each $x \in \mathcal{X}$. Because $\overline{\alpha}_{V^*}^{\kappa_s} \in \mathcal{K}$, we have $\overline{\alpha}_Y \in \mathcal{K}$. Recall from the lower bound that $\underline{\alpha}_Y := \underline{\alpha} \in \mathcal{K}$. Therefore, combining the lower and upper bounds of Y yields (17a).

Next, we show that (17b) holds along the solution $\{x_k^*\}_{k \in \mathbb{N}}$ to the DT closed-loop system in (2) generated by κ_γ^* . What follows holds for each $k \in \mathbb{N}$. By Lemma 3.3, V^* satisfies the Bellman equation, which evaluated at x_k^* yields

$$V^*(x_k^*) = R^*(x_k^*) + \gamma V^*(x_{k+1}^*).$$

Rearranging this equality to isolate $V^*(x_{k+1}^*)$ gives

$$V^*(x_{k+1}^*) = \frac{1}{\gamma} V^*(x_k^*) - \frac{1}{\gamma} R^*(x_k^*).$$

Subtracting $V^*(x_k^*)$ from both sides yields the change of V^* , namely,

$$\begin{aligned} V^*(x_{k+1}^*) - V^*(x_k^*) &= \frac{1}{\gamma} V^*(x_k^*) - \frac{1}{\gamma} R^*(x_k^*) - V^*(x_k^*) \\ &= \frac{1-\gamma}{\gamma} V^*(x_k^*) - \frac{1}{\gamma} R^*(x_k^*). \end{aligned}$$

Then, the change of Y is given by

$$\begin{aligned} Y(x_{k+1}^*) - Y(x_k^*) &= -(V^*(x_{k+1}^*) - V^*(x_k^*)) \\ &= \frac{1-\gamma}{\gamma} (-V^*(x_k^*)) + \frac{1}{\gamma} R^*(x_k^*) \end{aligned}$$

By item (i) in Assumption 3.6, $R^*(x_k^*) \leq -\underline{\alpha}(\sigma(x_k^*))$. Substituting this bound into the change of Y yields

$$Y(x_{k+1}^*) - Y(x_k^*) \leq \frac{1-\gamma}{\gamma} (-V^*(x_k^*)) - \frac{1}{\gamma} \underline{\alpha}(\sigma(x_k^*)). \quad (18)$$

Next, we bound both terms in the inequality above. First, by Lemma 3.8, $-V^*(x_k^*) \leq \overline{\alpha}_{V^*}^{\kappa_s}(\sigma(x_k^*))$. As $\mathcal{X}$ is compact and σ is continuous, σ attains its maximum on $\mathcal{X}$ by Theorem 4.16 in Rudin (1976). We define $\sigma_{\max} := \max_{x \in \mathcal{X}} \sigma(x)$. Because $\overline{\alpha}_{V^*}^{\kappa_s} \in \mathcal{K}$, $\overline{\alpha}_{V^*}^{\kappa_s}$ is strictly increasing. Hence, we can formulate the uniform upper bound

$$\frac{1-\gamma}{\gamma} (-V^*(x_k^*)) \leq \frac{1-\gamma}{\gamma} \overline{\alpha}_{V^*}^{\kappa_s}(\sigma(x_k^*)) \leq \frac{1-\gamma}{\gamma} \overline{\alpha}_{V^*}^{\kappa_s}(\sigma_{\max}) =: d_\gamma.$$

Next, as we showed that (17a) holds, $Y(x_k^*) \leq \overline{\alpha}_Y(\sigma(x_k^*))$. Because $\overline{\alpha}_Y \in \mathcal{K}$, its inverse $\overline{\alpha}_Y^{-1}$ exists and is strictly increasing, which implies $\sigma(x_k^*) \geq \overline{\alpha}_Y^{-1}(Y(x_k^*))$ and therefore $-\underline{\alpha}(\sigma(x_k^*)) \leq -\underline{\alpha}(\overline{\alpha}_Y^{-1}(Y(x_k^*)))$. Furthermore, because $\gamma \in (0, 1)$, it holds that $\frac{1}{\gamma} > 1$. Since $\underline{\alpha}$ takes non-negative values, this yields $-\frac{1}{\gamma} \underline{\alpha}(\sigma(x_k^*)) \leq -\underline{\alpha}(\sigma(x_k^*))$. Combining these inequalities gives

$$-\frac{1}{\gamma} \underline{\alpha}(\sigma(x_k^*)) \leq -\underline{\alpha}(\overline{\alpha}_Y^{-1}(Y(x_k^*))) =: -\alpha_Y(Y(x_k^*)),$$

where we define $\alpha_Y(s) := \underline{\alpha}(\overline{\alpha}_Y^{-1}(s))$ for each $s \in \mathbb{R}_{\geq 0}$. Because α_Y is the composition of class- $\mathcal{K}$ functions, it holds that $\alpha_Y \in \mathcal{K}$. Applying these bounds to (18) yields

$$Y(x_{k+1}^*) - Y(x_k^*) \leq -\alpha_Y(Y(x_k^*)) + d_\gamma.$$

Since Y satisfies (17a) and (17b), Y is a valid discrete-time ISS-Lyapunov function. By Lemma 3.5 in Jiang & Wang (2001), the DT closed-loop system is ISS with respect to the constant disturbance d_γ . Hence, the UPAS bound in (10) holds with $\nu(\gamma) := \nu_{\text{ISS}}(d_\gamma)$.

Lastly, we evaluate the limit of $\nu(\gamma)$ as $\gamma \rightarrow 1$. Recall that the constant disturbance is defined as

$$d_\gamma := \frac{1-\gamma}{\gamma} \bar{\alpha}_V^{\kappa_s}(\sigma_{\max}).$$

By item (ii) in Assumption 3.6, $\bar{\alpha}_V^{\kappa_s}$ is independent of γ . As shown above, σ attains its finite maximum on $\mathcal{X}$. Hence, $\bar{\alpha}_V^{\kappa_s}(\sigma_{\max})$ is a finite, non-negative constant independent of γ . Taking the limit as $\gamma \rightarrow 1$ gives

$$\lim_{\gamma \rightarrow 1} d_\gamma = \lim_{\gamma \rightarrow 1} \frac{1-\gamma}{\gamma} \bar{\alpha}_V^{\kappa_s}(\sigma_{\max}) = 0.$$

As $\nu_{\text{ISS}} \in \mathcal{K}$, it is continuous and zero at the origin, meaning $\nu_{\text{ISS}}(0) = 0$. Therefore, it holds that

$$\lim_{\gamma \rightarrow 1} \nu(\gamma) = \lim_{\gamma \rightarrow 1} \nu_{\text{ISS}}(d_\gamma) = \nu_{\text{ISS}}\left(\lim_{\gamma \rightarrow 1} d_\gamma\right) = \nu_{\text{ISS}}(0) = 0,$$

which concludes the proof. $\square$

Proof of Corollary 3.9.1. Pick any $\rho > 0$. By Theorem 3.9, there exist $\beta \in \mathcal{KL}$ and a continuous function $\nu: (0, 1) \rightarrow \mathbb{R}_{\geq 0}$ such that for each initial state $x_0^* \in \mathcal{X}$, the solution $\{x_k^*\}_{k \in \mathbb{N}}$ to the DT closed-loop system in (2) generated by κ_γ^* satisfies (10), where $\nu(\gamma) := \nu_{\text{ISS}}(d_\gamma)$ and $d_\gamma := \frac{1-\gamma}{\gamma} \bar{\alpha}_V^{\kappa_s}(\sigma_{\max})$ are taken from the proof of Theorem 3.9. To prove the bound in (11), we need to find $\gamma^* \in (0, 1)$ for which $\nu(\gamma) = \nu_{\text{ISS}}(d_\gamma) \leq \rho$ for each $\gamma \in [\gamma^*, 1)$.

As $\nu_{\text{ISS}} \in \mathcal{K}$, there exists a constant $\delta_\rho > 0$ such that $\nu_{\text{ISS}}(s) \leq \rho$ for each $s \in [0, \delta_\rho]$. Therefore, to find $\gamma^* \in (0, 1)$ that satisfies $\nu(\gamma) \leq \rho$ for each $\gamma \in [\gamma^*, 1)$, it is sufficient to show $d_\gamma \leq \delta_\rho$ for each $\gamma \in [\gamma^*, 1)$. If $\sigma_{\max} = 0$, then $d_\gamma = 0 \leq \delta_\rho$ holds trivially for any choice of $\gamma \in (0, 1)$. If $\sigma_{\max} > 0$, substituting the definition of d_γ and isolating γ gives

$$\frac{1-\gamma}{\gamma} \bar{\alpha}_V^{\kappa_s}(\sigma_{\max}) \leq \delta_\rho \implies \gamma \geq \frac{\bar{\alpha}_V^{\kappa_s}(\sigma_{\max})}{\delta_\rho + \bar{\alpha}_V^{\kappa_s}(\sigma_{\max})} =: \gamma^*.$$

Because $\bar{\alpha}_V^{\kappa_s}(\sigma_{\max})$ and δ_ρ are strictly positive, the denominator is strictly greater than the numerator, which guarantees $\gamma^* \in (0, 1)$. Finally, because the mapping $\gamma \mapsto \frac{1-\gamma}{\gamma}$ is strictly decreasing, any $\gamma \in [\gamma^*, 1)$ ensures $d_\gamma \leq d_{\gamma^*} \leq \delta_\rho$, and consequently $\nu(\gamma) \leq \rho$. This concludes the proof. $\square$

Proof of Lemma 4.2. Under Assumption 3.1, Theorem 3.2 guarantees that the mapping $\kappa \mapsto \mathcal{V}_\gamma(\cdot, \kappa)$ is continuous with respect to the supremum norm. Therefore, for the given tolerance $\delta > 0$, there exists $\varepsilon > 0$ such that for each $\tilde{\kappa} \in \mathbb{K}$ satisfying $\max_{x \in \mathcal{X}} |\kappa(x) - \tilde{\kappa}(x)| < \varepsilon$, it holds that $\max_{x \in \mathcal{X}} |\mathcal{V}_\gamma(x, \kappa) - \mathcal{V}_\gamma(x, \tilde{\kappa})| < \delta$.

Because $\mathbb{K}$ is the space of admissible continuous feedback laws, the given $\kappa \in \mathbb{K}$ satisfies $\kappa \in \mathcal{C}(\mathcal{X}, \mathcal{U})$. By Theorem 4.1, the class $\mathcal{NN}_{n,m,n+m+2}^\phi$ is dense in $\mathcal{C}(\mathcal{X}, \mathbb{R}^m)$ under the supremum norm. By the definition of a dense set, for the chosen $\varepsilon > 0$, there exists a network $\kappa_{\bar{\psi}} \in \mathcal{NN}_{n,m,n+m+2}^\phi$ satisfying $\max_{x \in \mathcal{X}} |\kappa(x) - \kappa_{\bar{\psi}}(x)| < \varepsilon$ (for example, see Definition 2.18 in Rudin (1976)).

By Definition 4.1, $\kappa_{\bar{\psi}}$ has a finite number of hidden layers. Consequently, it is parameterized by a weight vector $\bar{\psi} \in \mathbb{R}^\ell$ for some finite dimension $\ell \in \mathbb{N}_{>0}$. As a finite composition of continuous functions, $\kappa_{\bar{\psi}}$ is continuous. Restricting the domain and codomain of $\kappa_{\bar{\psi}}$ to $\mathcal{X}$ and $\mathcal{U}$, respectively, ensures $\kappa_{\bar{\psi}} \in \mathbb{K}$.

Because $\kappa_{\bar{\psi}} \in \mathbb{K}$ satisfies the required ε -bound, it immediately follows that $\max_{x \in \mathcal{X}} |\mathcal{V}_\gamma(x, \kappa) - \mathcal{V}_\gamma(x, \kappa_{\bar{\psi}})| \leq \delta$. Substituting $V_\gamma^\kappa(x) := \mathcal{V}_\gamma(x, \kappa)$ completes the proof. $\square$

Proof of Lemma 4.3. Under Assumption 3.1, Lemma 4.2 ensures there exist an integer $\ell \in \mathbb{N}_{>0}$ and a parameter vector $\bar{\psi} \in \mathbb{R}^\ell$ such that $\max_{x \in \mathcal{X}} |V_{\gamma}^{\kappa_{\gamma}^*}(x) - V_{\gamma}^{\kappa_{\bar{\psi}}}(x)| \leq \delta$. Hence, we obtain the following bound

$$\begin{aligned} \mathcal{J}_{\gamma}(\kappa_{\gamma}^*) - \mathcal{J}_{\gamma}(\kappa_{\bar{\psi}}) &= \int_{\mathcal{X}} (V_{\gamma}^{\kappa_{\gamma}^*}(x) - V_{\gamma}^{\kappa_{\bar{\psi}}}(x)) p_0(x) dx \\ &\leq \int_{\mathcal{X}} |V_{\gamma}^{\kappa_{\gamma}^*}(x) - V_{\gamma}^{\kappa_{\bar{\psi}}}(x)| p_0(x) dx \\ &\leq \int_{\mathcal{X}} \delta p_0(x) dx = \delta, \end{aligned}$$

and therefore $\mathcal{J}_{\gamma}(\kappa_{\bar{\psi}}) \geq \mathcal{J}_{\gamma}(\kappa_{\gamma}^*) - \delta$. Because $\bar{\psi}$ is a fixed vector in $\mathbb{R}^\ell$, we can choose a compact set $\Psi \subset \mathbb{R}^\ell$ large enough to contain $\bar{\psi}$. Under Assumption 3.1, Lemma 3.4 guarantees that for each $\gamma \in (0, 1)$, the functional $\kappa \mapsto \mathcal{J}_{\gamma}(\kappa)$ is continuous on $\mathbb{K}$ with respect to the supremum norm. Furthermore, because the given activation function ϕ is continuous under the conditions of Theorem 4.1, the mapping $\psi \mapsto \kappa_{\psi}$ is continuous with respect to the supremum norm. Consequently, the composite mapping $\psi \mapsto \mathcal{J}_{\gamma}(\kappa_{\psi})$ is continuous. Since Ψ is a compact subset of $\mathbb{R}^\ell$, this mapping attains its maximum on Ψ by Theorem 1 in Luenberger (1997, Section 2.13), which means $\psi^* \in \Psi$ exists. Because $\bar{\psi} \in \Psi$, this optimal solution trivially satisfies $\mathcal{J}_{\gamma}(\kappa_{\psi^*}) \geq \mathcal{J}_{\gamma}(\kappa_{\bar{\psi}}) \geq \mathcal{J}_{\gamma}(\kappa_{\gamma}^*) - \delta$. Furthermore, since κ_{γ}^* solves Problem 1, it is a global maximizer of $\mathcal{J}_{\gamma}$ over the entire admissible set $\mathbb{K}$. Because the parameterized feedback law inherently satisfies $\kappa_{\psi^*} \in \mathbb{K}$, it must hold that $\mathcal{J}_{\gamma}(\kappa_{\psi^*}) \leq \mathcal{J}_{\gamma}(\kappa_{\gamma}^*)$. Combining these bounds yields (14). $\square$

Proof of Lemma 4.4. For each $\kappa \in \mathbb{K}$, define the worst-case pointwise error $\mathcal{M}(\kappa) := \max_{x \in \mathcal{X}} (V_{\gamma}^{\kappa_{\gamma}^*}(x) - V_{\gamma}^{\kappa}(x))$ and the expected error $\mathcal{E}(\kappa) := \mathcal{J}_{\gamma}(\kappa_{\gamma}^*) - \mathcal{J}_{\gamma}(\kappa)$. Under Assumption 3.1, item (ii) in Assumption 3.6, and Assumption 3.7, Lemma 3.8 guarantees that $V_{\gamma}^{\kappa_{\gamma}^*}(x) \geq V_{\gamma}^{\kappa}(x)$ for each $(x, \kappa) \in \mathcal{X} \times \mathbb{K}$. Therefore, $\mathcal{M}(\kappa) \geq 0$ and $\mathcal{E}(\kappa) \geq 0$ for each $\kappa \in \mathbb{K}$. Under Assumption 3.1, the conditions of Theorem 3.2 and Lemma 3.4 are satisfied. In addition, because $\mathbb{K}$ is compact under the supremum norm, Theorem 3.2 and Lemma 3.4 guarantee $\mathcal{M}$ and $\mathcal{E}$ are continuous on $\mathbb{K}$. Since the admissible initial state distribution satisfies $p_0(x) > 0$ for each $x \in \mathcal{X}$, the expected error evaluates to zero if and only if the continuous pointwise error is identically zero; thus, $\mathcal{E}(\kappa) = 0 \iff \mathcal{M}(\kappa) = 0$.

To establish a lower bound on $\mathcal{E}$ in terms of $\mathcal{M}$, we define the function $\rho: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ as

$$\rho(s) := \min_{\tilde{\kappa} \in \mathbb{K}} (\mathcal{E}(\tilde{\kappa}) + \max\{0, s - \mathcal{M}(\tilde{\kappa})\}).$$

By evaluating ρ at $\mathcal{M}(\kappa)$ for a feedback law $\kappa \in \mathbb{K}$, we obtain the upper bound

$$\rho(\mathcal{M}(\kappa)) \leq \mathcal{E}(\kappa) + \max\{0, \mathcal{M}(\kappa) - \mathcal{M}(\kappa)\} = \mathcal{E}(\kappa).$$

Furthermore, because $\mathcal{M}(\tilde{\kappa}) \geq 0$ for each $\tilde{\kappa} \in \mathbb{K}$, evaluating ρ at $s = 0$ yields

$$\rho(0) = \min_{\tilde{\kappa} \in \mathbb{K}} \mathcal{E}(\tilde{\kappa}) = 0,$$

as $\mathcal{E}(\kappa_{\gamma}^*) = 0$. Next, for each fixed $\tilde{\kappa} \in \mathbb{K}$, the map $s \mapsto \max\{0, s - \mathcal{M}(\tilde{\kappa})\}$ is 1-Lipschitz continuous and non-decreasing in s . Because the pointwise infimum of any family of 1-Lipschitz functions is 1-Lipschitz (for example, see Proposition 2.3.9 in Cobzaş et al. (2019)), ρ is continuous. Since ρ is defined as the pointwise minimum of a family of non-decreasing functions, ρ is non-decreasing as well. Lastly, suppose for contradiction that $\rho(s) = 0$ for some $s > 0$. As ρ is continuous and $\mathbb{K}$ is a compact subset of a normed space, Theorem 1 in Luenberger (1997, Section 2.13) guarantees that there exists $\kappa_s \in \mathbb{K}$ at which the minimum of ρ is achieved. In particular, the minimum is obtained where both $\mathcal{E}(\kappa_s) = 0$ and $\max\{0, s - \mathcal{M}(\kappa_s)\} = 0$. However, $\mathcal{E}(\kappa_s) = 0 \implies \mathcal{M}(\kappa_s) = 0$. Substituting this yields

$$\rho(s) = \mathcal{E}(\kappa_s) + \max\{0, s - \mathcal{M}(\kappa_s)\} = \max\{0, s\} = s$$

which implies that $\rho(s) = s = 0$ and contradicts $s > 0$. Thus, $\rho(s) > 0$ for all $s > 0$.

Because ρ is continuous, non-decreasing, zero at the origin, and strictly positive for $s > 0$, by Lemma 4.3 in Khalil (2002), there exists $\underline{\alpha}_\gamma \in \mathcal{K}$ such that $\rho(s) \geq \underline{\alpha}_\gamma(s)$ for each $s \geq 0$. Therefore, $\mathcal{E}(\kappa) \geq \underline{\alpha}_\gamma(\mathcal{M}(\kappa))$ for each $\kappa \in \mathbb{K}$. Because $\underline{\alpha}_\gamma \in \mathcal{K}$, its inverse exists and $\underline{\alpha}_\gamma^{-1} \in \mathcal{K}$. Applying this inverse to both sides and defining $\alpha_\gamma := \underline{\alpha}_\gamma^{-1}$ yields $\mathcal{M}(\kappa) \leq \alpha_\gamma(\mathcal{E}(\kappa))$, completing the proof. $\square$

Proof of Theorem 4.5. As in the proof of Theorem 3.9, we establish the UPAS bound by adapting the DT ISS framework from Jiang & Wang (2001, Lemma 3.5). We treat the suboptimality induced by the discount factor $\gamma \in (0, 1)$ and the approximation error $\delta > 0$ as a constant external disturbance. Again, we map our setup to the generic perturbed DT system $x^+ = f(x, u)$ defined in Jiang & Wang (2001, Eq. (2)), where the disturbance input is a constant sequence satisfying $|u| \equiv \tilde{d}_{\gamma, \delta} \geq 0$.

For brevity, given $\gamma \in (0, 1)$ and $R: \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$, let $\hat{\kappa} := \kappa_{\psi^*}$ denote the optimal parameterized feedback law, and define $\hat{V}(x) := V_{\gamma}^{\hat{\kappa}}(x)$, $\hat{R}(x) := R(x, \hat{\kappa}(x))$, and $V^*(x) := V_{\gamma}^{\kappa^*}(x)$ for each $x \in \mathcal{X}$. We employ the same ISS-Lyapunov candidate used in Theorem 3.9, namely, $Y(x) := -V^*(x)$. Because Y evaluates V^* , the condition in (17a) is satisfied as shown in the proof of Theorem 3.9.

As Assumption 3.1 holds, Lemma 4.3 guarantees that for the given $\delta > 0$, the parameterized policy satisfies $\mathcal{J}_\gamma(\hat{\kappa}) \in [\mathcal{J}_\gamma(\kappa_\gamma^*) - \delta, \mathcal{J}_\gamma(\kappa_\gamma^*)]$. Furthermore, as Assumption 3.1, item (ii) in Assumption 3.6, and Assumption 3.7 hold, Lemma 4.4 implies that there exists $\alpha_\gamma \in \mathcal{K}$ such that (15) holds for each $\kappa \in \mathbb{K}$. Combining these results yields

$$\max_{x \in \mathcal{X}} \{V^*(x) - \hat{V}(x)\} \leq \alpha_\gamma(\mathcal{J}_\gamma(\kappa_\gamma^*) - \mathcal{J}_\gamma(\hat{\kappa})) \leq \alpha_\gamma(\delta) := \varepsilon_\delta.$$

Hence, $V^*(x) - \hat{V}(x) \leq \varepsilon_\delta$ for each $x \in \mathcal{X}$.

To evaluate the change of Y along the solutions of the DT closed-loop system in (2) generated by $\hat{\kappa}$, let $x \in \mathcal{X}$ and $x^+ = g(x, \hat{\kappa}(x))$. By Lemma 3.3, $\hat{V}$ satisfies the Bellman equation. Hence, $-\hat{V}(x^+) = -\gamma^{-1}\hat{V}(x) + \gamma^{-1}\hat{R}(x)$. By the optimality of κ_γ^* , $-V^*(x^+) \leq -\hat{V}(x^+)$. Next, we add $V^*(x)$ to both sides to obtain

$$\begin{aligned} -V^*(x^+) + V^*(x) &\leq -\hat{V}(x^+) + V^*(x) \\ &= -\frac{1}{\gamma}\hat{V}(x) + \frac{1}{\gamma}\hat{R}(x) + \underbrace{\frac{1}{\gamma}V^*(x) - \frac{1-\gamma}{\gamma}V^*(x)}_{=V^*(x)} \\ &= \frac{1-\gamma}{\gamma}(-V^*(x)) + \frac{1}{\gamma}(V^*(x) - \hat{V}(x)) + \frac{1}{\gamma}\hat{R}(x) \\ &\leq \frac{1-\gamma}{\gamma}(-V^*(x)) + \frac{\varepsilon_\delta}{\gamma} + \frac{1}{\gamma}\hat{R}(x). \end{aligned}$$

Following the procedure in the proof of Theorem 3.9, the change of Y evaluates to

$$Y(x^+) - Y(x) = -(V^*(x^+) - V^*(x)) \leq \frac{1-\gamma}{\gamma}(-V^*(x)) + \frac{\varepsilon_\delta}{\gamma} + \frac{1}{\gamma}\hat{R}(x).$$

By item (i) in Assumption 3.6, $\hat{R}(x) \leq -\underline{\alpha}(\sigma(x))$. Substituting this bound yields

$$Y(x^+) - Y(x) \leq \frac{1-\gamma}{\gamma}(-V^*(x)) - \frac{1}{\gamma}\underline{\alpha}(\sigma(x)) + \frac{\varepsilon_\delta}{\gamma}.$$

Because $\gamma \in (0, 1)$ and $\underline{\alpha}$ takes non-negative values, we apply the bound $-\frac{1}{\gamma}\underline{\alpha}(\sigma(x)) \leq -\underline{\alpha}(\sigma(x))$. By reusing the class- $\mathcal{K}$ -function bound α_Y and the constant bound $d_\gamma = \frac{1-\gamma}{\gamma}\bar{\alpha}_{V^*}^{\kappa_s}(\sigma_{\max})$ derived in Theorem 3.9, the inequality simplifies to

$$Y(x^+) - Y(x) \leq -\alpha_Y(Y(x)) + \tilde{d}_{\gamma, \delta},$$

Because the closed-loop system is ISS, Lemma 3.5 in [Jiang & Wang \(2001\)](#) guarantees the solutions $\{x_k\}_{k \in \mathbb{N}}$ generated by $\hat{\kappa}$ satisfy the bound

$$\sigma(x_k) \leq \max \left\{ \beta(\sigma(x_0), k), \nu_{\text{ISS}}(\tilde{d}_{\gamma, \delta}) \right\},$$

for some $\beta \in \mathcal{KL}$ and $\nu_{\text{ISS}} \in \mathcal{K}$. By defining the ultimate residual bound as $\tilde{\nu}(\gamma, \delta) := \nu_{\text{ISS}}(\tilde{d}_{\gamma, \delta})$, this recovers our target UPAS bound in (16).

Finally, for any fixed discount factor $\gamma \in (0, 1)$, taking the limit as the approximation error vanishes yields $\lim_{\delta \rightarrow 0} \epsilon_\delta = \lim_{\delta \rightarrow 0} \alpha_\gamma(\delta) = 0$, which results in $\lim_{\delta \rightarrow 0} \tilde{d}_{\gamma, \delta} = d_\gamma$. Because $\nu_{\text{ISS}} \in \mathcal{K}$ is continuous, this implies $\lim_{\delta \rightarrow 0} \tilde{\nu}(\gamma, \delta) = \nu_{\text{ISS}}(d_\gamma)$. Subsequently, as established in the proof of Theorem 3.9, taking the limit as the discount factor approaches one yields $\lim_{\gamma \rightarrow 1} \nu_{\text{ISS}}(d_\gamma) = 0$. This confirms the iterated limit $\lim_{\gamma \rightarrow 1} (\lim_{\delta \rightarrow 0} \tilde{\nu}(\gamma, \delta)) = 0$ and concludes the proof. $\square$